

Math.random()

The **Math.random()** function returns a floating-point, pseudo-random number that's greater than or equal to 0 and less than 1, with approximately uniform distribution over that range — which you can then scale to your desired range. The implementation selects the initial seed to the random number generation algorithm; it cannot be chosen or reset by the user.

Try it

JavaScript Demo: Math.random()

```
1 function getRandomInt(max) {  
2   return Math.floor(Math.random() * max);  
3 }  
4  
5 console.log(getRandomInt(3));  
6 // expected output: 0, 1 or 2  
7  
8 console.log(getRandomInt(1));  
9 // expected output: 0  
10  
11 console.log(Math.random());  
12 // expected output: a number from 0 to <1  
13
```

Run ›

Reset

Note: `Math.random()` does not provide cryptographically secure random numbers. Do not use them for anything related to security. Use the Web Crypto API instead, and more precisely the [window.crypto.getRandomValues\(\)](#) method.

Syntax

`Math.random()`

Return value

A floating-point, pseudo-random number between 0 (inclusive) and 1 (exclusive).

Examples

Note that as numbers in JavaScript are IEEE 754 floating point numbers with round-to-nearest-even behavior, the ranges claimed for the functions below (excluding the one for `Math.random()` itself) aren't exact. If extremely large bounds are chosen (2^{53} or higher), it's possible in *extremely* rare cases to reach the usually-excluded upper bound.

Getting a random number between 0 (inclusive) and 1 (exclusive)